

# Sequences and Series

1. Classify the sequence and state its common difference or ratio:  $7, -\frac{7}{2}, \frac{7}{4}, -\frac{7}{8}, \frac{7}{16}$ .
2. Write the next 3 terms:  $9, 11, 13, 15, \dots$
3. For the arithmetic sequence with  $a_1 = 7$  and  $d = -2$ , find  $a_{17}$ .
4. Find the common difference of  $-4, -8, -12, -16, \dots$
5. Find the sum of the first 21 terms of the arithmetic sequence with  $a_1 = 7$  and  $d = 8$ .
6. Find the missing term so the sequence is arithmetic:  $-6, -9, x, -15, -18$ .
7. Write the next 3 terms:  $8, 16, 32, 64, \dots$
8. For the geometric sequence with  $a_1 = 2$  and  $r = -2$ , find  $a_9$ .
9. Find the common ratio of  $6, 18, 54, 162, \dots$
10. Find the sum of the first 6 terms of the geometric sequence with  $a_1 = 6$  and  $r = 2$ .
11. Find the sum to infinity for the geometric series with  $a_1 = 8$  and  $r = \frac{1}{2}$ .
12. Given  $a_1 = -5$  and  $a_n = 2a_{n-1} - 4$ , find  $a_5$ .
13. A Fibonacci-type sequence begins  $0, 4$  and each new term is the sum of the previous two. Find  $a_8$ .
14. Evaluate  $\sum_{k=1}^6 (-k - 3)$ .
15. Determine whether  $a_n = \frac{4n^3 - 5n^2 - n - 1}{4}$  has a finite limit. If it does, find it; otherwise describe its behavior.
16. Determine whether  $a_n = (-\frac{1}{2})^n$  converges. If it does, give the limit.
17. Determine whether  $\sum_{n=1}^{\infty} \frac{1}{n^2}$  converges.
18. Determine whether the geometric series with  $a_1 = 5$  and  $r = -1$  converges; find its sum when possible.
19. Classify the convergence of  $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ .
20. Evaluate the telescoping sum  $\sum_{k=1}^{17} (\frac{1}{k} - \frac{1}{k+1})$ .

## Solutions

1. The sequence is geometric; its common ratio is  $-\frac{1}{2}$ .
2. The common difference is 2, so the next terms are 17, 19, 21.
3.  $a_n = 7 + (-2)(n - 1)$ , so  $a_{17} = -25$ .
4.  $d = a_2 - a_1 = -8 - (-4) = -4$ .
5.  $S_{21} = \frac{21}{2}(7 + 167) = 1827$ .
6.  $x = -12$ .
7. The common ratio is 2, so the next terms are 128, 256, 512.
8.  $a_n = 2 \cdot (-2)^{n-1}$ , so  $a_9 = 512$ .
9.  $r = \frac{a_2}{a_1} = 3$ .
10.  $S_6 = a_1 \frac{1-r^6}{1-r} = 378$ .
11.  $S_\infty = \frac{a_1}{1-r} = 16$ .
12. The generated terms are  $-5, -14, -32, -68, -140$ , so  $a_5 = -140$ .
13. Continuing gives 0, 4, 4, 8, 12, 20, 32, 52, so the requested term is 52.
14. The terms are  $-4, -5, -6, -7, -8, -9$ ; their sum is  $-39$ .
15. There is no finite limit. The quotient behaves like  $n^3$  and tends to  $+\infty$ .
16. It converges to 0.
17. This p-series converges because  $p = 2$  is greater than 1.
18. It diverges because  $|r| \geq 1$ .
19. It converges absolutely; the absolute p-series has  $p=2>1$ .
20. All intermediate terms cancel, leaving  $1 - \frac{1}{18} = \frac{17}{18}$ .

## Polynomial Limit Cases

21. Determine whether  $a_n = \frac{4n+3}{6n^3+2n^2+9n+1}$  has a finite limit. If it does, find it; otherwise describe its behavior.
22. Evaluate  $\lim_{n \rightarrow \infty} \frac{6n^2+3n+6}{9n^2+1}$ , or state that it does not exist.
23. Evaluate  $\lim_{n \rightarrow \infty} \frac{7n^4+3n^3-6n^2+8n+9}{2n^3+8n^2+4n+7}$ , or state that it does not exist.
24. Determine whether  $a_n = \frac{-8n^4-8n^3-7n^2+5n-8}{6n}$  has a finite limit. If it does, find it; otherwise describe its behavior.
25. Determine whether  $a_n = (-1)^n \frac{3n+9}{-3n-6}$  has a finite limit. If it does, find it; otherwise describe its behavior.

## Solutions

21. The denominator has higher degree, so the limit is 0.
22. The degrees are equal, so the limit is the ratio of the leading coefficients:  $\frac{2}{3}$ .
23. There is no finite limit. The quotient behaves like  $\frac{7}{2}n$  and tends to  $+\infty$ .
24. There is no finite limit. The quotient behaves like  $-\frac{4}{3}n^3$  and tends to  $-\infty$ .
25. The polynomial quotient tends to  $-1$ , but  $(-1)^n$  makes the sequence oscillate. The limit does not exist.

## Deterministic Closed-form Limits

26. Evaluate  $\lim_{x \rightarrow -2} -2x^3 + 2x^2 - 2x + 3$ .

27. Evaluate  $\lim_{x \rightarrow 0} \frac{3x^2 - 4x - 4}{4x^2 + 4x + 6}$ .

28. Evaluate  $\lim_{x \rightarrow -3} \sqrt{-3x - 8}$ .

29. Evaluate  $\lim_{x \rightarrow -4} e^{2x + 6}$ .

30. Evaluate  $\lim_{x \rightarrow 1} \ln(x + 6)$ .

31. Evaluate  $\lim_{x \rightarrow -\pi} \sin(x)$ .

32. Evaluate  $\lim_{x \rightarrow \frac{2\pi}{3}} \cos(x)$ .

33. Evaluate  $\lim_{x \rightarrow -4} |5x + 3|$ .

34. Evaluate  $\lim_{n \rightarrow \infty} \left(1 + \frac{3}{n}\right)^{-4n}$ .

35. Evaluate  $\lim_{x \rightarrow -3} \frac{6x^2 + 17x - 3}{-4x^2 - 20x - 24}$ .

36. Evaluate  $\lim_{x \rightarrow 0} \frac{\sin(-2x)}{-5x}$ .

37. Evaluate  $\lim_{x \rightarrow 0} \frac{\tan(6x)}{-5x}$ .

38. Evaluate  $\lim_{x \rightarrow 0} \frac{1 - \cos(5x)}{5x^2}$ .

## Solutions

26. The function is continuous and defined at the given point, so direct substitution gives 31.
27. The function is continuous and defined at the given point, so direct substitution gives  $-\frac{2}{3}$ .
28. The function is continuous and defined at the given point, so direct substitution gives 1.
29. The function is continuous and defined at the given point, so direct substitution gives  $e^{-2}$ .
30. The function is continuous and defined at the given point, so direct substitution gives  $\ln(7)$ .
31. The function is continuous and defined at the given point, so direct substitution gives 0.
32. The function is continuous and defined at the given point, so direct substitution gives  $-\frac{1}{2}$ .
33. The function is continuous and defined at the given point, so direct substitution gives 17.
34. Using  $\lim_{n \rightarrow \infty} (1 + u/n)^n = e^u$ , the exponent in the result is  $-12$ , so the limit is  $e^{-12}$ .
35. Cancel the common factor  $x + 3$  and substitute  $x = -3$  into  $\frac{6x-1}{-4x-8}$  to obtain  $-\frac{19}{4}$ .
36. Apply  $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$  and account for the scale factors to obtain  $\frac{2}{5}$ .
37. Apply  $\lim_{u \rightarrow 0} \frac{\tan u}{u} = 1$  and account for the scale factors to obtain  $-\frac{6}{5}$ .
38. Apply  $\lim_{u \rightarrow 0} \frac{1 - \cos u}{u^2} = \frac{1}{2}$  and account for the scale factors to obtain  $\frac{5}{2}$ .